

ERW: Carbon Dioxide Removal

**A technical derivation of the per-pixel CDR calculation in the
ex-ante ERW pipeline, with sources for every mechanism
and constant**

[Bisrat Haile](#)
CIMMYT

Last updated: July 10, 2026

Contents

About this report	1
1 Overview — the calculation as one chain	2
1.1 Symbols and default values	3
2 Feedstock chemistry — the Ca and Mg content	4
3 The stoichiometric CDR ceiling	5
4 The reactive (effective) fraction	7
5 The grain-size factor	8
6 The climate and soil-pH factor	9
7 The pedogenic-carbonate deduction	11
8 Assembling the per-tonne and per-hectare gross removal	13
9 The net-export correction — the load-bearing accounting step	14
10 Regime dependence and time-resolved crediting	16
11 Worked example (illustrative pixel)	17
12 Assumptions and limitations	18
13 Provenance of every number and formula	20
13.1 Standard physical constants	20
13.2 In-model derivations and modelling choices (provenance = code line)	21
13.3 Literature-anchored values (source, exact locus, confidence) .	22
References	24

List of Tables

About this report

This report documents exactly how the pipeline turns a tonne of applied basalt into a quantity of durably removed carbon dioxide. It walks the full chain implemented in `erw/erw-3-basalt-requirements.R` (feedstock chemistry and the per-tonne potential) and `erw/erw-9-cdr-yield.R` (the spatial climate, soil, and accounting terms), states each governing equation, gives the numerical default, and cites the peer-reviewed source that grounds each assumption and mechanism.

Every source is cited in APA author–year form and resolves in the **References**. Where a value is a modelling choice rather than a measured constant, this is stated plainly and the choice is defended against the cited range.

Chapter 1

Overview — the calculation as one chain

Durable CDR (tonnes CO₂ per hectare) at a cropland pixel is built in five conceptual moves, then corrected by a sixth:

1. **Stoichiometric ceiling** — the maximum CO₂ a tonne of basalt could remove, fixed by its Ca and Mg oxide content (pure chemistry).
2. **Reactive fraction** — only a fraction of that ceiling actually weathers on a decadal horizon.
3. **Grain-size factor** — finer grinding exposes more surface area and speeds reaction.
4. **Climate & soil-pH factor** — temperature, moisture, and soil acidity rescale the reaction rate relative to a temperate reference site.
5. **Pedogenic-carbonate deduction** — in dry soils some alkalinity re-precipitates as carbonate locally, re-releasing CO₂.
6. **Net-export correction** — alkalinity consumed neutralizing the soil's own exchangeable acidity re-releases its CO₂ (as agricultural lime does) and is *not* durable removal; only alkalinity exported beyond the acidity demand counts.

Assembled, the per-hectare durable removal is

$$\text{CDR}_{\text{net}}(x) = \max\left(\underbrace{D(x) \cdot \frac{1}{1000} \Phi(x) C_{\text{max}} (1 - p_{\text{ped}}(x))}_{\text{gross removal}} - \underbrace{F \cdot S(x)}_{\text{acidity re-release}}, 0\right)$$

where D is the basalt application rate (t ha⁻¹), C_{max} the stoichiometric ceiling (kg CO₂ t⁻¹), Φ the effective reactive fraction (grain × climate × pH), p_{ped} the

pedogenic-carbonate fraction, F the CO₂ re-released per tonne of acidity neutralized, and S the soil's acidity expressed as a lime requirement (t CaCO₃-eq ha⁻¹). Each term is derived below.

1.1 Symbols and default values

Symbol	Meaning	Default	Units	Set in
$f_{\text{CaO}}, f_{\text{MgO}}$	basalt oxide mass fractions	0.10 / 0.07	–	erw-3
C_{max}	stoichiometric CDR ceiling	309.8	kg CO ₂ / t basalt	erw-3
ϕ_0	reference reactive fraction	0.20	–	erw-3
d	grain size	50	μm	erw-3
g	grain-size factor	1.414	–	erw-3
$\phi_{\text{ref}} = \phi_0 g$	reference reactive fraction (grain-adjusted)	0.283	–	erw-9
$T_{\text{ref}}, P_{\text{ref}}$	climate reference anchors	11 / 1000	°C / mm	erw-9
Φ	effective reactive fraction (capped at 1)	per pixel	–	erw-9
p_{ped}	pedogenic-carbonate fraction	0–0.90	–	erw-9
F	CO ₂ re-released per t CaCO ₃ -eq neutralized	0.88	t CO ₂ / t CaCO ₃	erw-9
S	soil acidity as lime requirement	per pixel	t CaCO ₃ / ha	erw-3

Chapter 2

Feedstock chemistry — the Ca and Mg content

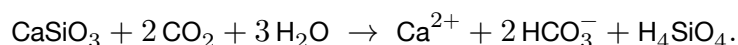
What the model assumes. A representative tropical-belt basalt fine with $f_{\text{CaO}} = 0.10$ and $f_{\text{MgO}} = 0.07$ by mass (erw-3, the basalt list). Minor oxides (K_2O 1.5 %, Na_2O 2.5 %, P_2O_5 0.3 %) and trace metals (Ni 100 ppm, Cr 200 ppm) are carried for nutrient and risk screening but do not enter the CDR term.

Why these values. Basalts carry their weathering “fuel” as $\text{CaO} \approx 8\text{--}12$ % and $\text{MgO} \approx 5\text{--}10$ % by mass. For calibration, the six commercial ERW basalts assayed by [Lewis et al. \(2021\)](#) (their Table 6, p. 10) report *elemental* Ca = 4.08–6.72 % and Mg = 1.30–5.86 %, i.e. $\text{CaO} \approx 5.7\text{--}9.4$ % and $\text{MgO} \approx 2.2\text{--}9.7$ % — so the model’s 10 % CaO sits at/just above the top of that particular set and 7 % MgO mid-range, consistent with the broader basalt population ([Renforth, 2012](#), Abstract p. 229). The pair is a representative default, exposed as an override once quarry-specific assays are available.

Chapter 3

The stoichiometric CDR ceiling

Mechanism. When a Ca- or Mg-silicate weathers under a CO₂-charged soil solution, each mole of divalent base cation released is charge-balanced by two moles of bicarbonate:



So the ceiling is **2 mol CO₂ per mol of Ca²⁺ or Mg²⁺**. Using molar masses (CaO 56.08, MgO 40.30, CO₂ 44.01 g mol⁻¹):

$$1 \text{ g CaO} \rightarrow \frac{2 \times 44.01}{56.08} = 1.570 \text{ g CO}_2, \quad 1 \text{ g MgO} \rightarrow \frac{2 \times 44.01}{40.30} = 2.185 \text{ g CO}_2.$$

Per tonne of basalt,

$$C_{\max} = (f_{\text{CaO}} \cdot 1.570 + f_{\text{MgO}} \cdot 2.185) \times 1000 = (0.10 \cdot 1.570 + 0.07 \cdot 2.185) \times 1000 \approx \mathbf{309.8 \text{ kg CO}_2 \text{ t}^{-1}}.$$

Source. The 1:2 cation-to-CO₂ bicarbonate stoichiometry is the mechanistic basis of ERW CDR accounting ([Hartmann et al., 2013](#)). The cation-based “maximum uptake” calculation from oxide chemistry, and the resulting **≈ 0.3 t CO₂ per t basalt** for basic rocks, trace to [Renforth \(2012\)](#) and are used at scale by [Beerling et al. \(2020\)](#) and [Strefler, Amann, Bauer, Kriegler, and Hartmann \(2018\)](#). It is an **upper bound** in two senses: it assumes complete dissolution, and it credits the full 2 mol CO₂ per cation, which only holds while carbon is

carried as dissolved bicarbonate — on downstream carbonate precipitation ($\text{Ca}^{2+} + 2 \text{HCO}_3^- \rightarrow \text{CaCO}_3 + \text{CO}_2 + \text{H}_2\text{O}$) half is re-released, so the long-term net can fall toward ~1 mol CO_2 per mol Ca ([Renforth, 2019](#)). Both reductions are applied below (pedogenic deduction, §[7](#); net-export, §[9](#)).

Chapter 4

The reactive (effective) fraction

What the model assumes. Only $\phi_0 = 0.20$ of the stoichiometric ceiling is realized on the accreditation horizon at the reference grain size and climate (erw-3, reactive_fraction_ref).

Why. Field and mesocosm studies consistently report realized CDR far below the stoichiometric ceiling because dissolution is incomplete on decadal timescales, secondary minerals sequester cations, and cation exchange delays export. [Lewis et al. \(2021\)](#) report CDR of 1.3–8.5 t CO₂ ha⁻¹ over 15 yr at 50 t ha⁻¹ — i.e. **26–170 kg CO₂ per tonne basalt actually realized**, against the ~310 kg ceiling. Their per-basalt spread (Cragmill ≈ 9 %, Blue Ridge ≈ 21 %, Tichum ≈ 57 % of stoichiometric over 15 yr) brackets $\phi_0 = 0.20$ as a central value. Basalt field trials ([Kantola et al., 2023](#); [Larkin et al., 2022](#)) and the uncertainty analyses of [Reershemius and Suhrhoff \(2024\)](#) sit within this band. A cool-temperate reality check sits at the low end: [Dupla, Bertagni, and Grand \(2025\)](#) found detectable dissolution but CO₂-removal rates < 10 % of some other studies over ~1000 days of Swiss vineyard trials. The model therefore treats $\phi_0 = 0.20$ as a defensible central anchor (not a single measured constant), later scaled up in warm/wet climates via the climate factor and swept in erw-8.

Chapter 5

The grain-size factor

Mechanism. Dissolution is surface-reaction-limited, so rate scales with specific surface area, which for a given mass scales inversely with particle diameter. The model uses

$$g = \left(\frac{d_{\text{ref}}}{d} \right)^{\beta}, \quad d_{\text{ref}} = 100 \mu\text{m}, \quad \beta = 0.5,$$

giving $g = (100/50)^{0.5} = \mathbf{1.414}$ at the 50 μm default.

Why $\beta = 0.5$. Empirically, silt-dominated basalt fines (<45 μm) weather at roughly twice the rate of sand-dominated material (150–500 μm), implying $\beta \approx 0.35$ –0.5. The square-root value is the parameter-free midpoint (rate $\propto \sqrt{\text{surface-area}}$) and avoids introducing an extra free constant. Grain-size control on weathering rate and the associated grinding-energy penalty are quantified by [Strefler et al. \(2018\)](#) ($E \approx 0.07 \text{ GJ t}^{-1}$ at 50 μm rising to $\approx 3 \text{ GJ t}^{-1}$ at 2 μm) and used in the ERW potential estimates of [Beerling et al. \(2020\)](#). Grain size is the model’s dominant decision variable and is swept 10–500 μm in erw–8.

Reference reactive fraction after grinding. erw–9 combines the two:

$$\phi_{\text{ref}} = \phi_0 g = 0.20 \times 1.414 = \mathbf{0.283}.$$

At the reference climate this yields an effective per-tonne removal of $\phi_{\text{ref}} C_{\text{max}} = 0.283 \times 309.8 \approx \mathbf{87.6 \text{ kg CO}_2 \text{ t}^{-1}}$ (CDR_eff_kg_per_t_ref), consistent with the [Lewis et al. \(2021\)](#) mesocosm midpoint.

Chapter 6

The climate and soil-pH factor

The reference reactive fraction is anchored to a temperate-humid site (US Corn Belt, $T_{\text{ref}} = 11\text{ }^{\circ}\text{C}$, $P_{\text{ref}} = 1000\text{ mm}$). A dimensionless factor rescales it per pixel:

$$\Phi(x) = \min\left(\phi_{\text{ref}} \cdot f_{\text{MAT}} \cdot f_{\text{MAP}} \cdot f_{\text{pH}}, 1\right),$$

$$f_{\text{MAT}} = \text{clamp}\left(\frac{\text{MAT}}{11}, 0.3, 3.0\right), \quad f_{\text{MAP}} = \text{clamp}\left(\frac{\text{MAP}}{1000}, 0.3, 3.0\right).$$

Temperature term. Silicate dissolution is thermally activated (Arrhenius), so warmer soils weather faster; the linear-in- T multiplier is a first-order surrogate for that acceleration. Basalt-specific dissolution experiments give the activation energies directly: [Gudbrandsson, Wolff-Boenisch, Gíslason, and Oelkers \(2011\)](#) measured crystalline basalt over 5–75 °C, and [Gíslason and Oelkers \(2003\)](#) basaltic glass over 6–150 °C, both confirming strong Arrhenius acceleration; [Palandri and Kharaka \(2004\)](#) tabulate the activation energies used in reactive-transport ERW models. The general temperature control on weathering flux is established by [White and Brantley \(2003\)](#) and [Brantley, Kubicki, and White \(2008\)](#), and the climate control on ERW CDR capacity specifically is mapped by [Baek et al. \(2023\)](#).

Moisture term. Weathering export requires water moving through the profile; CDR flux rises with precipitation and drainage. The canonical field demonstration is [White and Blum \(1995\)](#), who show Si and Na weathering fluxes rising systematically with runoff across watersheds; the dependence underpins the global ERW capacity maps of [Baek et al. \(2023\)](#) and [Beerling et al. \(2020\)](#).

pH term (acid catalysis). Silicate dissolution is acid-catalyzed: the rate rises as pH falls (higher H^+ activity), so acidic soils dissolve basalt faster. The standard three-mechanism rate law, $r = k_{\text{acid}} a_{H^+}^n + k_{\text{neutral}} + k_{\text{base}} a_{H^+}^{-m}$, is compiled by [Palandri and Kharaka \(2004\)](#); basalt-specific pH data ([Gíslason & Oelkers, 2003](#); [Gudbrandsson et al., 2011](#)) show dissolution rate rising sharply below neutral pH. The model uses a triangular factor peaking near pH 5:

$$f_{\text{pH}} = \begin{cases} 0.5 & \text{pH} < 4 \\ 0.5 + 0.5(\text{pH} - 4) & 4 \leq \text{pH} < 5 \\ 1.0 - 0.25(\text{pH} - 5) & 5 \leq \text{pH} < 7 \\ 0.5 & \text{pH} \geq 7 \end{cases}$$

The rise from pH 4 $\rightarrow$ 5 and the $\sim 2\times$ suppression on calcareous soils (pH > 7) follow the acid-catalyzed dissolution kinetics of [White and Brantley \(2003\)](#) and [Brantley et al. \(2008\)](#). The peak near pH 5 and the roll-off below it is a modelling compromise: true rate is monotone in H^+ , but very low pH in these soils co-occurs with Al toxicity and low base status, so the factor is capped. This whole factor is flagged in code as a **v1 empirical proxy** to be replaced by a reactive-transport surrogate ([Kanzaki, Planavsky, & Reinhard, 2023](#)) when calibration data allow.

Chapter 7

The pedogenic-carbonate deduction

Mechanism. In semi-arid and arid soils a substantial fraction of the Ca/Mg alkalinity released by weathering precipitates *locally* as secondary (pedogenic) $\text{CaCO}_3/\text{MgCO}_3$ instead of exporting as bicarbonate. Carbonate precipitation releases roughly half the CO_2 the silicate originally consumed ($\text{Ca}^{2+} + 2 \text{HCO}_3^- \rightarrow \text{CaCO}_3 + \text{CO}_2 + \text{H}_2\text{O}$), so this alkalinity does not deliver durable removal and must be deducted:

$$\text{exported fraction} = 1 - p_{\text{ped}}(x).$$

Keying on aridity. p_{ped} is binned on the Aridity Index ($\text{AI} = \text{MAP} / \text{PET}$) using [United Nations Environment Programme \(1992\)](#) classes:

$\text{AI} = \text{MAP}/\text{PET}$	Class	p_{ped}
< 0.05	Hyperarid	0.90
0.05–0.20	Arid	0.70
0.20–0.50	Semi-arid	0.40
0.50–0.65	Dry sub-humid	0.15
> 0.65	Humid	0.00

When the CGIAR-CSI Global Aridity Index raster ([Zomer, Xu, & Trabucco, 2022](#)) is present the deduction is keyed directly on AI; otherwise erw-9 falls back to a MAP-only proxy with the same class boundaries, which over-deducts in cool highlands (Ethiopia, Kenya, Rwanda) where low PET keeps true AI humid.

Sources. [Renforth \(2019\)](#) quantifies the “ ≈ 0.5 mol CO₂ re-released per mol CaCO₃ precipitated” penalty; [Harrington, Hilton, and Henderson \(2023\)](#) measure a concrete 16–27 % CDR reduction from carbonate precipitation during riverine transport; [Kanzaki et al. \(2023\)](#) treat it as a permanence-loss term; and the [Beerling et al. \(2020\)](#) supplementary accounting flags it. AI bands and dataset: [United Nations Environment Programme \(1992\)](#); [Zomer et al. \(2022\)](#).

Open question. Whether ERW-formed pedogenic carbonate is durable ($\sim 10^4$ yr) or transient is contested: fertilizer-derived strong acids can re-dissolve it and re-release the CO₂, in which case it should not be credited as storage at all. The exact loss fraction and its aridity scaling are among the least-constrained numbers in the whole calculation — the aridity bins here are a deliberately coarse v1 proxy, and reviewers will (rightly) push on the loss coefficients.

Chapter 8

Assembling the per-tonne and per-hectare gross removal

Per tonne of basalt at pixel x :

$$c(x) = \Phi(x) C_{\max} (1 - p_{\text{ped}}(x)) \quad [\text{kg CO}_2 \text{ t}^{-1}],$$

and the **gross** per-hectare removal for an application rate $D(x)$ (t ha⁻¹):

$$\text{CDR}_{\text{gross}}(x) = D(x) \cdot \frac{c(x)}{1000} \quad [\text{t CO}_2 \text{ ha}^{-1}].$$

$D(x)$ comes from erw-3: either the agronomically **targeted** lime-equivalent rate (basalt-to-lime ratio $1/\text{effective_NV} = 1/0.0996 \approx 10.04$ t basalt per t CaCO₃-eq, applied to the LiTAS lime requirement) or a **uniform** 10/20/50 t ha⁻¹ counterfactual. The targeted rate uses the LiTAS lime-requirement method of [Aramburu Merlos, Silva, Baudron, and Hijmans \(2023\)](#); uniform rates match the modelling convention of [Beerling et al. \(2020\)](#), [Baek et al. \(2023\)](#), and [Kantola et al. \(2023\)](#).

Chapter 9

The net-export correction — the load-bearing accounting step

This is the single most consequential step and the main methodological update in this version of the pipeline.

The problem. The gross removal above credits carbon for *all* weathered alkalinity. But on an acid soil, alkalinity has two competing fates drawing on one pool:

- **Neutralize exchangeable acidity** (displace $\text{Al}^{3+}/\text{H}^+$, raise pH). This is the agronomic benefit — but neutralizing acidity consumes the bicarbonate and **releases the CO_2 back to the atmosphere** ($\text{HCO}_3^- + \text{H}^+ \rightarrow \text{H}_2\text{O} + \text{CO}_2$, with the protons supplied by Al^{3+} hydrolysis). No net carbon is removed. This is exactly why agricultural lime is *not* a CDR technology even though it raises yield the same way.
- **Export as bicarbonate** to rivers and the ocean. This is durable removal ($>10^4$ yr).

Counting the acidity-consumed tranche as CDR double-counts the same alkalinity — once as yield benefit, once as removal.

The fix. Deduct the acidity sink, a mass balance:

$$\text{CDR}_{\text{net}}(x) = \max(\text{CDR}_{\text{gross}}(x) - F \cdot S(x), 0)$$

- $S(x)$ = the soil's acidity as a lime requirement ($\text{t CaCO}_3\text{-eq ha}^{-1}$), already computed by erw-3 (Kamprath / LiTAS). **Regime-dependent:**

year-1/NPV use the *standing* exchangeable acidity (large); equilibrium uses only the *annual re-acidification* increment (small).

- F = CO₂ re-released per tonne of CaCO₃-equivalent acidity neutralized. It is **not a tuned parameter** — it falls out of the feedstock's own constants:

$$F = \frac{\text{CDR}_{\text{eff}}}{\text{effective_NV} \times 1000} = \frac{87.6}{99.6} \approx \mathbf{0.88 \text{ t CO}_2 (\text{t CaCO}_3)^{-1}}.$$

Consequences. Because standing acidity is large, the year-1 deduction removes ~80 % of credited CDR; at equilibrium only the small annual re-acidification is deducted, so most removal survives. Durable CDR is therefore largely a **steady-state (maintenance)** phenomenon, and year-1 ERW on acid soils is, to first order, *liming with a small carbon tail*.

Sources. The carbon-fate rule — bicarbonate produced against *strong/mineral* acid (including exchangeable soil acidity) is a net CO₂ source, only bicarbonate produced against *carbonic* acid is a sink — is established in the agricultural-liming carbon-accounting literature ([Hamilton, Kurzman, Arango, Jin, & Robertson, 2007](#); [West & McBride, 2005](#)). The most direct ERW statement of the mechanism is [Dietzen and Rosing \(2023\)](#), whose protocol states that non-carbonic acids must be accounted for whenever soil pH (H₂O) is below **6.3**: as pH falls, mineral dissolution rises but CO₂-capture *efficiency* falls because H⁺/exchangeable acidity consumes the released alkalinity, so naïve cation-based accounting over-credits CDR on acid soils. That only *exported* alkalinity is durable removal — and that cation-release mass balance is an upper bound needing a downstream-loss correction — is the central message of the ERW MRV literature ([Kanzaki et al., 2023](#); [Levy et al., 2024](#); [Reershemius et al., 2023](#); [Suhrhoff et al., 2024](#); [Zhang et al., 2022](#)).

One caveat, stated plainly. This is a first-order *sequential* bound: it assumes the acidity sink is filled before any alkalinity exports. In reality weathering, neutralization, and leaching proceed simultaneously, so some alkalinity exports even on acid soils, and re-acidification continually regenerates the sink. The truth lies between the gross (“all exports”) and net (“sink filled first”) bounds; the pipeline writes both `cdr_yield_*` (gross) and `cdr_yield_*_netexport` (net) rasters and leads the carbon case on the equilibrium regime, where the two converge.

Chapter 10

Regime dependence and time-resolved crediting

The same per-hectare removal is credited under three temporal regimes (in erw-7):

- **Year-1** — one up-front application; CDR accrues over a 5-year phasing (30/25/20/15/10 %, ~80 % by year 5), first-order-kinetics-shaped ([Kanzaki et al., 2023](#); [Beerling et al. \(2020\)](#) SI).
- **NPV @ 10 %** — the 5-year CDR tail is discounted by $\text{CDR_NPV_FACTOR} = \sum_t f_t / (1 + r)^t / \sum_t f_t \approx 0.79$.
- **Equilibrium** — steady-state maintenance dose only; the net-export sink is the small annual re-acidification, so durable removal is largest here.

Chapter 11

Worked example (illustrative pixel)

A warm, wet, moderately acid cropland pixel: MAT = 25 °C, MAP = 1200 mm, soil pH = 6.0, humid ($p_{\text{ped}} = 0$), targeted rate $D = 15 \text{ t ha}^{-1}$, standing acidity $S_{\text{std}} = 3.0 \text{ t CaCO}_3 \text{ ha}^{-1}$, maintenance $S_{\text{maint}} = 0.30 \text{ t CaCO}_3 \text{ ha}^{-1}$.

Step	Computation	Result
f_{MAT}	clamp(25/11)	2.27
f_{MAP}	clamp(1200/1000)	1.20
f_{pH}	$1 - 0.25(6 - 5)$	0.75
Φ	$\min(0.283 \cdot 2.27 \cdot 1.20 \cdot 0.75, 1)$	0.578
c (per t)	$0.578 \cdot 309.8 \cdot (1 - 0)$	179 kg t ⁻¹
Gross CDR	$15 \cdot 179 / 1000$	2.69 t ha ⁻¹
Net (year-1)	$\max(2.69 - 0.88 \cdot 3.0, 0)$	0.05 t ha⁻¹
Net (equilibrium)	$\max(2.69 - 0.88 \cdot 0.30, 0)$	2.43 t ha⁻¹

The collapse from 2.69 → 0.05 t ha⁻¹ in year-1, and its survival at equilibrium, is the net-export effect in miniature.

Chapter 12

Assumptions and limitations

#	Assumption / mechanism	Status	Upgrade path
1	10 %/7 % CaO/MgO basalt	Representative; override per quarry	Quarry assays via erw-basalt-access
2	1:2 cation:CO ₂ stoichiometry	Well-established chemistry	—
3	0.20 reference reactive fraction	Central of a wide (0.08–0.55) field range	Site calibration as trials accrue
4	$\beta = 0.5$ grain-size exponent	Defensible midpoint (0.35–0.5)	Feedstock-specific SSA-rate data
5	Linear climate multipliers	v1 empirical proxy	Reactive-transport surrogate (Kanzaki et al., 2023)
6	Triangular pH factor	v1 proxy; true rate monotone in H ⁺	Kinetic dissolution model
7	Aridity-binned pedogenic loss	Coarse; MAP proxy over-deducts highlands	CGIAR-CSI AI raster (Zomer et al., 2022)

8	Net-export sequential bound	First-order; presented with gross bound	Coupled transport- reaction
---	--------------------------------	---	-----------------------------------

The two quantities the downstream economics is most sensitive to — the realized reactive fraction (Chapter 4) and the pedogenic/net-export loss (Chapters 7 and 9) — are also the least empirically resolved, which is why they are carried explicitly as bounds and swept in erw-8 rather than fixed silently.

Chapter 13

Provenance of every number and formula

Each quantity in this report traces to one of three kinds of source: a **standard physical constant**, an **in-model derivation or modelling choice** (whose provenance is a specific line in the pipeline code, not a journal page), or a **literature-anchored value** (traced to an exact page / table / figure / equation, with a confidence flag from the source-verification pass). Line numbers refer to the scripts as committed. Confidence: **V** = verified from full text, **A** = confirmed from the paper’s abstract only (full text paywalled), **C** = modelling choice / convention rather than a single measured value.

13.1 Standard physical constants

Quantity	Value	Source and locus
Molar mass CaO / MgO	56.08 / 40.30 g mol ⁻¹	IUPAC/CIAAW standard atomic weights (Prohaska et al., 2022) (Ca 40.078, Mg 24.305, O 15.999)
Molar mass CO ₂ / CaCO ₃	44.01 / 100.09 g mol ⁻¹	Prohaska et al. (2022) (C 12.011)
1 mol Ca ²⁺ /Mg ²⁺ → 2 mol CO ₂ (bicarbonate basis)	stoichiometry	Hartmann et al. (2013) , Eq. (8), §2.1 (forsterite reaction) — V ; Beerling et al. (2020) , Methods “CDR”, Eqs (4)–(5)

13.2 In-model derivations and modelling choices (provenance = code line)

Quantity	Value	Code locus
CaO / MgO mass fraction	0.10 / 0.07	erw-3.R:55–56 (choice; §13.3)
Grain size d	50 μm	erw-3.R:67 (decision variable)
Reference reactive fraction ϕ_0	0.20	erw-3.R:74 (anchored; §13.3)
Theoretical CCE (1.783, 2.481 g/g)	0.352	erw-3.R:82–84 (molar-mass arithmetic)
Stoichiometric ceiling C_{max} (1.570, 2.185 g/g)	309.8 kg t^{-1}	erw-3.R:87–91
Grain-size factor $g = (100/d)^{0.5}$	1.414	erw-3.R:100–102 ($\beta = 0.5$ choice; §13.3)
Grinding curve $E = 0.07(50/d)^{1.2}$	—	erw-3.R:113–118 (anchored to Strefler; §13.3)
Effective NV	0.0996 t t^{-1}	erw-3.R:130
Basalt-to-lime ratio 1/NV	10.04	erw-3.R:131
Reference effective CDR $\phi_{\text{ref}} = \phi_0 g$	87.6 kg t^{-1}	erw-3.R:132
$T_{\text{ref}}, P_{\text{ref}}$	0.283	erw-9.R:48
$f_{\text{MAT}}, f_{\text{MAP}}$ clamp [0.3, 3.0]	11 $^{\circ}\text{C}$, 1000 mm	erw-9.R:83–84 (US Corn Belt normal)
f_{pH} triangular, peak ≈ 5	—	erw-9.R:86–87
climate_factor = $f_{\text{MAT}} f_{\text{MAP}} f_{\text{pH}}$	—	erw-9.R:90–93
Pedogenic fractions 0.90/0.70/0.40/0.15/0	—	erw-9.R:96
Reactive fraction cap at 1	—	erw-9.R:152–163 (bands from §13.3)
Per-tonne CDR $c(x)$	—	erw-9.R:181
CDR yield $= D \cdot c/1000$	—	erw-9.R:186
$F = \text{CDR}_{\text{eff}}/(\text{NV} \times 1000)$	—	erw-9.R:194
Net-export $\max(\text{gross} - FS, 0)$	0.88	erw-9.R:233
	—	erw-9.R:240–244

Quantity	Value	Code locus
CDR phasing 30/25/20/15/10 %	—	erw-7.R:106 (shape from §13.3)
Discount rate	10 %	erw-7.R:100
CDR NPV factor	≈ 0.79	erw-7.R:109–111

13.3 Literature-anchored values (source, exact locus, confidence)

Quantity	Source — exact locus	Conf.
Representative basalt CaO/MgO	Lewis et al. (2021) , Table 6, p. 10 (elemental Ca 4.08–6.72 %, Mg 1.30–5.86 %); Renforth (2012) , Abstract, p. 229	V / A
Stoichiometric max $\approx$ 0.3 t CO ₂ t ⁻¹ (basic rock)	Renforth (2012) , Abstract, p. 229 ; Lewis et al. (2021) , Table 6 p. 10 (R_{CO_2} (CaO+MgO) = 0.13–0.30), formula Eq. (5), p. 13	A / V
Reactive fraction ~ 0.20 (realized/stoichiometric)	Lewis et al. (2021) , Table 5, p. 5 (15-yr CDR 1.3–8.5 t ha ⁻¹ at 50 t ha ⁻¹ : Cragmill 1.3 $\rightarrow$ Tichum 8.5; ≈ 9 –57 % of stoichiometric); Abstract p. 1 ; §4.1 p. 13	V
Low-end realized fraction (< 10 %)	Dupla et al. (2025) , Abstract	A
Grinding energy 0.07 GJ t ⁻¹ @ 50 μm ; 3 GJ t ⁻¹ @ 2 μm	Strefler et al. (2018) , “Mining, crushing and grinding” subsection (~p. 7); Fig. 1 (lower axis); SI C, Eq. C1	V
Grain-size $\rightarrow$ dissolution-rate scaling ($\beta \approx 0.35$ –0.5)	Strefler et al. (2018) , Eq. (3) & Fig. 1 (~p. 3); Eqs (1)–(2), SI B (β is a modelling convention grounded in surface-area kinetics)	V / C

Quantity	Source — exact locus	Conf.
Temperature (Arrhenius) dependence	Gudbrandsson et al. (2011) ; Gíslason and Oelkers (2003) ; Palandri and Kharaka (2004) (activation-energy tables)	V
Moisture / runoff dependence	White and Blum (1995)	V
pH acid-catalysis rate law	Palandri and Kharaka (2004) (three-mechanism rate law); Gíslason and Oelkers (2003) ; White and Brantley (2003)	V
Aridity bands 0.05/0.20/0.50/0.65	United Nations Environment Programme (1992) , reproduced verbatim in IPCC SRCCL (Mirzabaev et al., 2019), Ch. 3 §3.1.1, Fig. 3.1, p. ~255 (note: Zomer et al. (2022) uses a 0.03 hyperarid variant — not used here)	V
Pedogenic carbonate re-releases ≈ 0.5 mol CO ₂ per mol CaCO ₃	Renforth (2019) , Eq. (3), Introduction ; Beerling et al. (2020) , Methods “CDR”, Eq. (5) (1 mol CO ₂ /mol Ca vs 1.72 via ocean)	V
Carbonate-precipitation CDR loss 16–27 %	Harrington et al. (2023) , Abstract	A
Acidity-consumed alkalinity $\neq$ CDR; pH (H ₂ O) 6.3 threshold	Dietzen and Rosing (2023) , Abstract	A
Ocean-transport permanence / leakage (~9 % silicate)	Kanzaki et al. (2023) , Results/Discussion, Fig. 2	V
LiTAS lime-requirement method (targeted rate)	Aramburu Merlos et al. (2023) , §4.1.6, Eqs (11)–(12) ($a = 0.60$, $b = 0.92$)	V

Quantity	Source — exact locus	Conf.
Uniform rates 10/20/50 t ha ⁻¹	Beerling et al. (2020) baseline 40 t ha⁻¹ yr⁻¹ (Methods “Baseline simulations”); Baek et al. (2023) 10 t ha ⁻¹ ; Kantola et al. (2023) 50 t ha ⁻¹	A
CDR phasing shape (first-order, ~80 % by yr 5)	Kanzaki et al. (2023) ; Beerling et al. (2020) SI	C

Note on two thresholds. (i) The pipeline’s hyperarid break is 0.05 (UNEP 1992 / IPCC SRCCL; [Mirzabaev et al., 2019](#); [United Nations Environment Programme, 1992](#)), *not* the 0.03 in [Zomer et al. \(2022\)](#); if the CGIAR-CSI raster is loaded its own scaling is respected but the class breaks stay at the code values. (ii) [Renforth \(2012\)](#)’s “~0.3 t CO₂ t⁻¹” and its cation-uptake formula could only be verified at the abstract; the in-text derivation is behind a paywall and is therefore marked **A**.

*References here match the master catalogue in docs/erw-parameters.md, extended with additional sources added and DOI-verified for this report to source the weathering-kinetics, stoichiometry, net-export, and constant-provenance items specifically. Page/table/equation loci in §13.3 were located by direct inspection of each source; items marked **A** were confirmable only at the abstract because the full text is paywalled.*

References

- Aramburu Merlos, F., Silva, J. V., Baudron, F., & Hijmans, R. J. (2023). Estimating lime requirements for tropical soils: Model comparison and development. *Geoderma*, 432, 116421. (<https://doi.org/10.1016/j.geoderma.2023.116421>)
- Baek, S. H., Kanzaki, Y., Lora, J. M., Planavsky, N., Reinhard, C. T., & Zhang, S. (2023). Impact of climate on the global capacity for enhanced rock weathering on croplands. *Earth's Future*, 11, e2023EF003698. (<https://doi.org/10.1029/2023EF003698>)
- Beerling, D. J., Kantzas, E. P., Lomas, M. R., Wade, P., Eufrazio, R. M., Renforth, P., ... others (2020). Potential for large-scale CO₂ removal via enhanced rock weathering with croplands. *Nature*, 583, 242–248. (<https://doi.org/10.1038/s41586-020-2448-9>)
- Brantley, S. L., Kubicki, J. D., & White, A. F. (2008). Kinetics of water-rock interaction. In S. L. Brantley, J. D. Kubicki, & A. F. White (Eds.), *Kinetics of water-rock interaction*. New York: Springer. (<https://doi.org/10.1007/978-0-387-73563-4>)
- Dietzen, C., & Rosing, M. T. (2023). Quantification of CO₂ uptake by enhanced weathering of silicate minerals applied to acidic soils. *International Journal of Greenhouse Gas Control*, 125, 103872. (<https://doi.org/10.1016/j.ijggc.2023.103872>)
- Dupla, X., Bertagni, M. B., & Grand, S. (2025). Three years of field trials indicate a sustained enhanced rock weathering signal with limited CO₂ removal. *Environmental Science & Technology*, 59(48), 25751–25764. (<https://doi.org/10.1021/acs.est.5c09820>)
- Gíslason, S. R., & Oelkers, E. H. (2003). Mechanism, rates, and consequences of basaltic glass dissolution: II. an experimental study of the dissolution rates of basaltic glass as a function of pH and temperature. *Geochimica et Cosmochimica Acta*, 67(20), 3817–3832. ([https://doi.org/10.1016/S0016-7037\(03\)00176-5](https://doi.org/10.1016/S0016-7037(03)00176-5))
- Gudbrandsson, S., Wolff-Boenisch, D., Gíslason, S. R., & Oelkers, E. H. (2011). An experimental study of crystalline basalt dissolution from $2 \leq \text{pH} \leq 11$ and temperatures from 5 to 75 °C. *Geochimica et Cosmochimica Acta*, 75(19), 5496–5509. (<https://doi.org/10.1016/j.gca.2011.06.035>)

- Hamilton, S. K., Kurzman, A. L., Arango, C., Jin, L., & Robertson, G. P. (2007). Evidence for carbon sequestration by agricultural liming. *Global Biogeochemical Cycles*, 21, GB2021. (<https://doi.org/10.1029/2006GB002738>)
- Harrington, K. J., Hilton, R. G., & Henderson, G. M. (2023). Implications of the riverine response to enhanced weathering for CO₂ removal in the UK. *Applied Geochemistry*, 152, 105643. (<https://doi.org/10.1016/j.apgeochem.2023.105643>)
- Hartmann, J., West, A. J., Renforth, P., Köhler, P., De La Rocha, C. L., Wolf-Gladrow, D. A., ... Scheffran, J. (2013). Enhanced chemical weathering as a geoengineering strategy to reduce atmospheric carbon dioxide, supply nutrients, and mitigate ocean acidification. *Reviews of Geophysics*, 51(2), 113–149. (<https://doi.org/10.1002/rog.20004>)
- Kantola, I. B., Blanc-Betes, E., Masters, M. D., Chang, E., Marklein, A., Moore, C. E., ... DeLucia, E. H. (2023). Improved net carbon budgets in the U.S. midwest through direct measured impacts of enhanced weathering. *Global Change Biology*, 29(24), 7012–7028. (<https://doi.org/10.1111/gcb.16903>)
- Kanzaki, Y., Planavsky, N. J., & Reinhard, C. T. (2023). New estimates of the storage permanence and ocean co-benefits of enhanced rock weathering. *PNAS Nexus*, 2(4), pgad059. (<https://doi.org/10.1093/pnasnexus/pgad059>)
- Larkin, C. S., Andrews, M. G., Pearce, C. R., Yeong, K. L., Beerling, D. J., Bellamy, J., ... James, R. H. (2022). Quantification of CO₂ removal in a large-scale enhanced weathering field trial on an oil palm plantation in sabah, malaysia. *Frontiers in Climate*, 4, 959229. (<https://doi.org/10.3389/fclim.2022.959229>)
- Levy, C. R., Almaraz, M., Beerling, D. J., Raymond, P., Reinhard, C. T., Suhrhoff, T. J., & Taylor, L. (2024). Enhanced rock weathering for carbon removal—monitoring and mitigating potential environmental impacts on agricultural land. *Environmental Science & Technology*, 58(39), 17215–17226. (<https://doi.org/10.1021/acs.est.4c02368>)
- Lewis, A. L., Sarkar, B., Wade, P., Kemp, S. J., Hodson, M. E., Taylor, L. L., ... others (2021). Effects of mineralogy, chemistry and physical properties of basalts on carbon capture potential and plant-nutrient element release via enhanced weathering. *Applied Geochemistry*, 132, 105023. (<https://doi.org/10.1016/j.apgeochem.2021.105023>)
- Mirzabaev, A., Wu, J., Evans, J., García-Oliva, F., Hussein, I. A. G., Iqbal, M. H., ... Weltz, M. (2019). Desertification. In P. R. Shukla, J. Skea, et al. (Eds.), *Climate change and land: An IPCC special report (SRCCL)* (chap. 3). Intergovernmental Panel on Climate Change. (<https://www.ipcc.ch/srccl/chapter/chapter-3/>)
- Palandri, J. L., & Kharaka, Y. K. (2004). *A compilation of rate parameters of water–mineral interaction kinetics for application to geochemical modeling* (Open-File Report No. 2004-1068). U.S. Geological Survey. (<https://pubs.usgs.gov/publication/ofr20041068>)

- Prohaska, T., Irrgeher, J., Benefield, J., Bohlke, J. K., Chesson, L. A., Coplen, T. B., ... others (2022). Standard atomic weights of the elements 2021 (IUPAC technical report). *Pure and Applied Chemistry*, 94(5), 573–600. (<https://doi.org/10.1515/pac-2019-0603>)
- Reershemius, T., Kelland, M. E., Jordan, J. S., Davis, I. T., D’Ascanio, R., Kalderon-Asael, B., ... Planavsky, N. J. (2023). Initial validation of a soil-based mass-balance approach for empirical monitoring of enhanced rock weathering rates. *Environmental Science & Technology*, 57(48), 19497–19507. (<https://doi.org/10.1021/acs.est.3c03609>)
- Reershemius, T., & Suhrhoff, T. J. (2024). On error, uncertainty, and assumptions in calculating carbon dioxide removal rates by enhanced rock weathering. *Global Change Biology*, 30(1), e17025. (<https://doi.org/10.1111/gcb.17025>)
- Renforth, P. (2012). The potential of enhanced weathering in the UK. *International Journal of Greenhouse Gas Control*, 10, 229–243. (<https://doi.org/10.1016/j.ijggc.2012.06.011>)
- Renforth, P. (2019). The negative emission potential of alkaline materials. *Nature Communications*, 10, 1401. (<https://doi.org/10.1038/s41467-019-09475-5>)
- Strefler, J., Amann, T., Bauer, N., Kriegler, E., & Hartmann, J. (2018). Potential and costs of carbon dioxide removal by enhanced weathering of rocks. *Environmental Research Letters*, 13(3), 034010. (<https://doi.org/10.1088/1748-9326/aaa9c4>)
- Suhrhoff, T. J., Reershemius, T., Wang, J., Jordan, J. S., Reinhard, C. T., & Planavsky, N. J. (2024). Measuring enhanced weathering: inorganic carbon-based approaches may be required to complement cation-based approaches. *Frontiers in Climate*, 6, 1352825. (<https://doi.org/10.3389/fclim.2024.1352825>)
- United Nations Environment Programme. (1992). *World atlas of desertification*. London: Edward Arnold.
- West, T. O., & McBride, A. C. (2005). The contribution of agricultural lime to CO₂ emissions in the united states: dissolution, transport, and net emissions. *Agriculture, Ecosystems & Environment*, 108(2), 145–154. (<https://doi.org/10.1016/j.agee.2005.01.002>)
- White, A. F., & Blum, A. E. (1995). Effects of climate on chemical weathering in watersheds. *Geochimica et Cosmochimica Acta*, 59(9), 1729–1747. ([https://doi.org/10.1016/0016-7037\(95\)00078-E](https://doi.org/10.1016/0016-7037(95)00078-E))
- White, A. F., & Brantley, S. L. (2003). The effect of time on the weathering of silicate minerals: why do weathering rates differ in the laboratory and field? *Chemical Geology*, 202(3–4), 479–506. (<https://doi.org/10.1016/j.chemgeo.2003.03.001>)
- Zhang, S., Planavsky, N. J., Katchinoff, J., Raymond, P. A., Kanzaki, Y., Reershemius, T., & Reinhard, C. T. (2022). River chemistry constraints on the carbon capture potential of surficial enhanced rock weathering. *Limnology & Oceanography*, 67(S1), S148–S157. (<https://doi.org/10.1002/lno.12244>)

- Zomer, R. J., Xu, J., & Trabucco, A. (2022). Version 3 of the global aridity index and potential evapotranspiration database. *Scientific Data*, 9, 409. (<https://doi.org/10.1038/s41597-022-01493-1>)